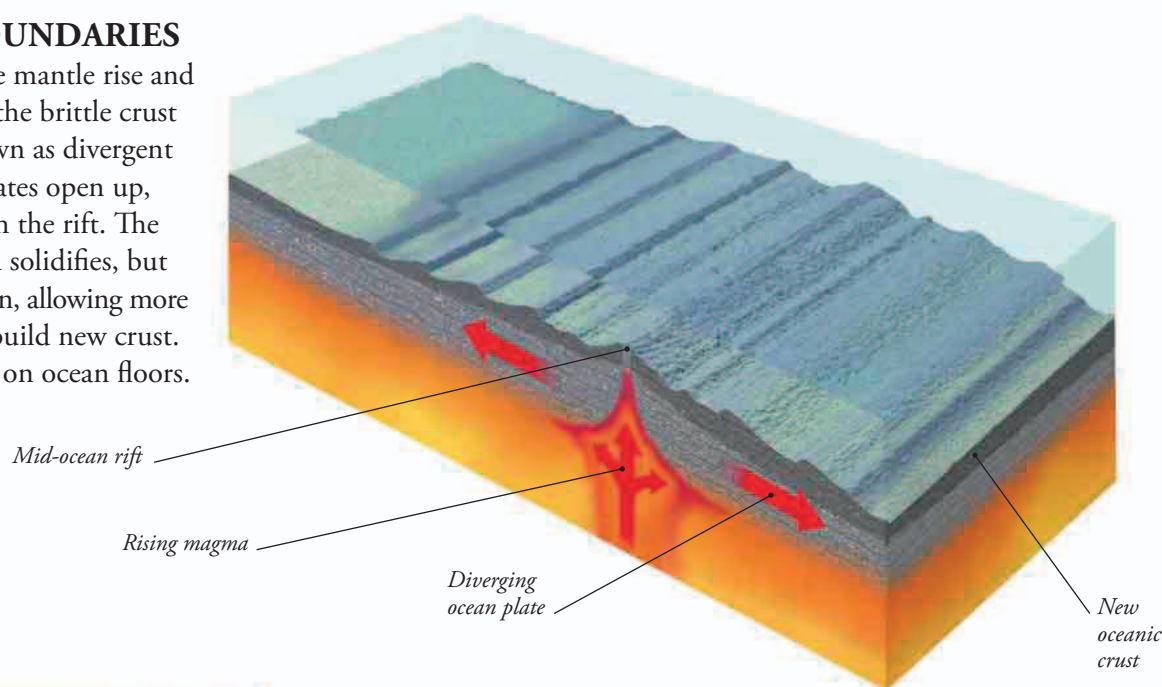


Plate boundaries

Since tectonic plates are pulling apart in some places and pushing together in others, different types of boundaries are created between the plates. In places where the plates are ripping apart, new crust is created as molten rock wells up from below. Where they are grinding together, old crust is destroyed. But there are also boundaries where slabs of crust simply slide past each other.

DIVERGENT BOUNDARIES

As heat currents in the mantle rise and divide, they rip apart the brittle crust at spreading rifts known as divergent boundaries. Where plates open up, magma erupts through the rift. The molten rock cools and solidifies, but then the rift opens again, allowing more magma to erupt and build new crust. Most of these rifts are on ocean floors.



EXPANDING OCEANS

When rifts open, they form valleys that tend to fill with water. In North Africa, this process has created the Red Sea (left). The Atlantic looked like this about 160 million years ago. It has been expanding ever since from the rift of the Mid-Atlantic Ridge at the rate of about 1 in (25 mm) per year.

◀ SPREADING SEA

The Red Sea has formed in the rift where the African Plate and the Arabian Plate are being pulled apart. It is getting steadily wider and will eventually become an ocean.

FACT!

Many rift valleys are filled with long, deep lakes. Among these is Lake Baikal in Russia, which is the deepest lake on Earth and contains a fifth of the world's fresh water.